

PYTHON MACHINE LEARNING PORTFOLIO

Predicting Daily Rainfall in Melbourne Using Machine Learning

A Temporal Validation and Model Comparison Study

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Companion technical artifact: fully executed Jupyter notebook

Executive Summary

This portfolio study develops an earlier IBM Machine Learning Certificate rainfall-classification exercise into a **research-oriented machine-learning analysis**. The central question is **whether meteorological information available on day t can predict rainfall on day $t+1$ in the Melbourne metropolitan area**, and **how alternative classifiers perform when evaluation respects the temporal order of the data**.

The redesign focuses on empirical validity rather than headline accuracy. The analysis retains 8,443 Melbourne-area observations with an observed target, uses pipeline-contained imputation rather than complete-case deletion, partitions data chronologically into development (<2016), validation (2016), and final test (2017) periods, and tunes hyperparameters with expanding-year temporal cross-validation. **Logistic Regression, Random Forest, and Histogram Gradient Boosting** are compared against a **naïve prior baseline** using **accuracy, balanced accuracy, rainfall precision/recall/F1, ROC-AUC, Average Precision, and Brier score**.

Histogram Gradient Boosting was selected on 2016 validation performance. With a validation-selected probability threshold of 0.438, the final 2017 test achieved **85.7% accuracy**, 0.676 balanced accuracy, 71.7% rainfall precision, 38.8% rainfall recall, 0.503 rainfall F1, **0.841 ROC-AUC**, 0.611 Average Precision, and a Brier score of 0.109. The model therefore discriminated rainy from dry days substantially better than chance, but its 2017 sensitivity to rainfall was materially weaker than its validation-year sensitivity. This deterioration is analytically important: it illustrates why temporally realistic validation and class-specific metrics matter more than a single random-split accuracy statistic.

Held-out permutation importance identified **Pressure3pm, Humidity3pm, WindGustSpeed, and Sunshine** as the inputs on which generalization performance relied most strongly. Station-level performance was comparatively stable, whereas seasonal error analysis exposed a notable weakness in the small 2017 winter subset, where the chosen threshold detected none of the 11 rainy cases. A station-holdout robustness check further showed that generalization to an unseen nearby station remained useful but uneven.

For a business-analytics audience, the project demonstrates a transferable **predictive decision workflow**: define the forecast horizon, control leakage, build reproducible preprocessing pipelines, select models under realistic validation, evaluate minority events, optimize decision thresholds, inspect calibration and feature reliance, and diagnose heterogeneous errors. These practices translate directly to applications such as churn, credit risk, fraud, demand forecasting, operational disruption, and response modeling.

Portfolio Contribution at a Glance

Dimension	Evidence in this study
Python / data engineering	pandas data audit, datetime handling, feature engineering, reproducible scikit-learn pipelines
Machine learning	Logistic Regression, Random Forest, Histogram Gradient Boosting, GridSearchCV
Validation design	Chronological holdout plus expanding-year temporal cross-validation
Imbalanced classification	Average Precision, class-specific precision/recall/F1, balanced accuracy
Decision analytics	Validation-only threshold selection and illustrative asymmetric error cost

Model interpretation	Held-out permutation importance, calibration, station/season error diagnostics
Research practice	Explicit provenance, limitations, reproducibility, and separation of predictive from causal claims

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1. Introduction and Research Motivation

Predictive machine learning is most valuable when the evaluation design reflects the decision environment in which a model would actually be used. **Weather prediction** provides a useful test bed for this principle because observations are explicitly ordered in time, rainfall is an imbalanced event, and measurement availability constrains what information can legitimately be used at prediction time.

This project originated as a rainfall-classification exercise in the IBM Machine Learning Certificate. The portfolio redevelopment does not present that educational exercise as independent meteorological research. Instead, it uses the original technical foundation—Python data manipulation, heterogeneous preprocessing, scikit-learn pipelines, GridSearchCV, Random Forest, Logistic Regression, classification diagnostics, and feature importance—as the starting point for a more rigorous predictive study.

The research question is: **How effectively can meteorological observations available on day t predict rainfall on day $t+1$ in the Melbourne metropolitan area, and how do alternative machine-learning classifiers perform under temporally realistic out-of-sample evaluation?**

The methodological emphasis is aligned with research reporting practice: clearly specify the prediction target, separate model development from final evaluation, document preprocessing, benchmark multiple models, report more than one performance metric, and discuss limitations rather than treating the best score as the sole result. Published guidance on machine-learning predictive modeling similarly stresses sufficient reporting of model specification, validation, and performance so that readers can assess model validity.

2. Data and Analytical Scope

2.1 Dataset

The source file contains 145,460 daily observations and 23 variables from Australian weather stations, including temperature, rainfall, evaporation, sunshine, wind, humidity, pressure, cloud cover, location, RainToday, and RainTomorrow. The portfolio analysis focuses on **Melbourne**, **Melbourne Airport**, and **Watsonia**, yielding 8,443 observations with usable dates and observed **RainTomorrow** outcomes from 1 July 2008 through 25 June 2017.

The Australian Bureau of Meteorology states that daily rainfall is generally observed at 9 a.m. local time and represents precipitation received during the preceding 24 hours. This measurement convention reinforces the need to state the forecasting horizon carefully rather than loosely relabeling daily variables.

Sample component	Observations / rate
Raw Australian dataset	145,460 rows
Melbourne-metro analytical sample	8,443 rows
No rain next day	6,416 (75.99%)
Rain next day	2,027 (24.01%)
Development period (<2016)	6,945; rain rate 23.87%
Validation period (2016)	972; rain rate 27.88%
Final test period (2017)	526; rain rate 18.63%

2.2 Missing Data

Missing data are substantial for several predictors: Sunshine is missing in 48.01% of raw records, Evaporation in 43.17%, Cloud3pm in 40.81%, and Cloud9am in 38.42%. The original certificate notebook used complete-case deletion, reducing the full dataset from 145,460 to 56,420 rows before geographic filtering. The portfolio version instead removes only observations without a usable date or target and handles predictor missingness inside the fitted pipeline. Numerical variables are median-imputed; categorical variables use the most frequent category.

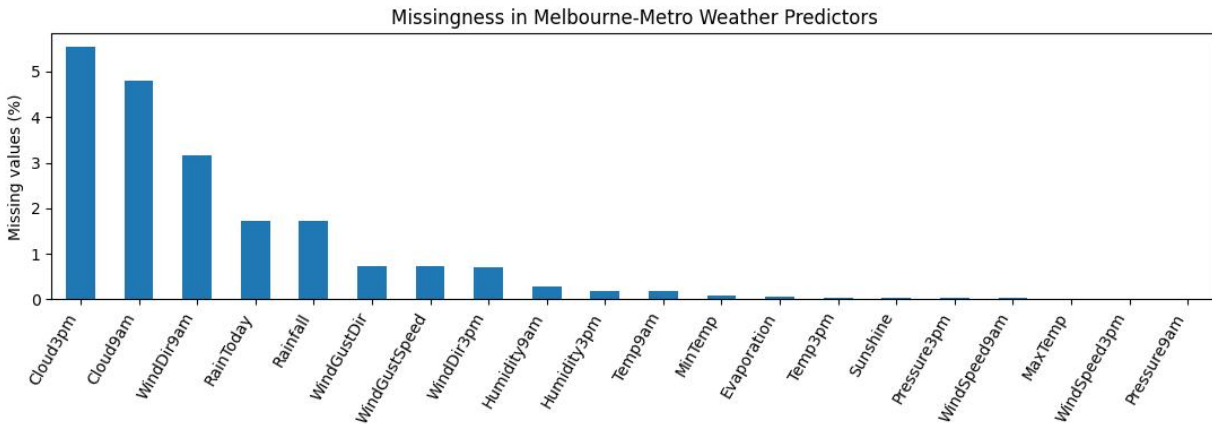


Figure 1. Missing data profile for predictors in the Melbourne-metro analytical sample.

2.3 Descriptive Patterns

Rain prevalence varies meaningfully over time. It reached 27.88% in 2016 but fell to 18.63% in the partial 2017 test period. This shift matters because threshold performance and class-specific error rates depend partly on event prevalence and temporal conditions.

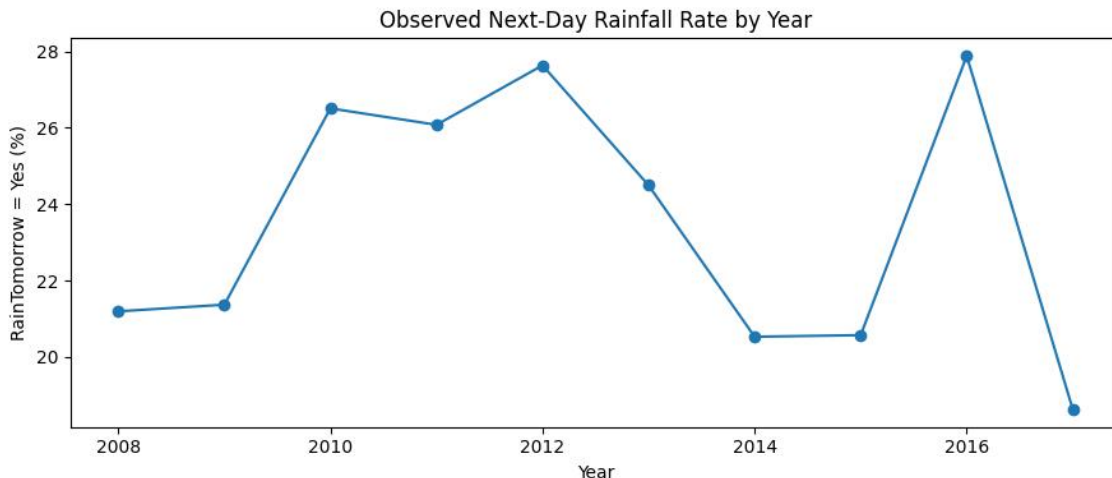


Figure 2. Observed next-day rainfall rate by calendar year.

Seasonality is also visible. Across the full Melbourne-metro sample, next-day rain occurred on 29.31% of winter observations and 27.62% of spring observations, compared with 16.59% in summer. Rain prevalence also differed modestly across stations: 26.12% in Melbourne, 24.61% in Watsonia, and 21.70% at Melbourne Airport.

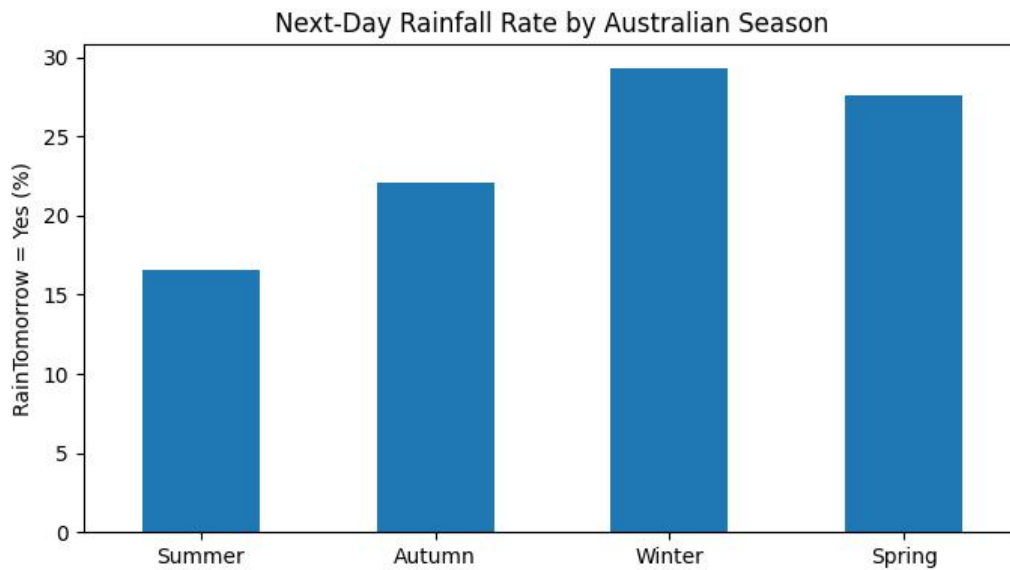


Figure 3. Next-day rainfall prevalence by Australian season.

3. Methodology

3.1 Forecasting Horizon and Feature Engineering

The outcome target is **RainTomorrow**. Each row is interpreted as information available for day t used to predict whether rainfall occurs on day $t+1$. This preserves the temporal meaning of the original dataset and makes the deployment assumption explicit.

The model matrix contains 24 predictors: 18 numerical and 6 categorical. The date itself is excluded as a direct predictor, while annual periodicity is represented through sine and cosine transformations of day-of-year. An interpretable Southern Hemisphere Season feature is also retained. The cyclical representation avoids treating the end and beginning of the calendar year as artificially distant.

3.2 Leakage-Safe Preprocessing

All data-dependent preprocessing is contained within a scikit-learn **Pipeline** and **ColumnTransformer**. Numerical features undergo median imputation and standardization. Categorical features undergo mode imputation and one-hot encoding with unseen categories ignored. Keeping these operations inside the fitted pipeline ensures that imputation, scaling, and encoding parameters are learned only from the relevant training partition.

3.3 Temporal Validation Design

The central methodological change is the replacement of randomized splitting with chronological evaluation. Referring from Scikit-learn, ordinary cross-validation can be inappropriate for time-ordered data because it can train on future observations and evaluate on past observations. The present study therefore uses observations **before 2016 for model development, 2016 for validation and threshold selection, and 2017 as an untouched final test**.

Hyperparameters are tuned with six expanding-year folds. Each fold trains on all eligible earlier years and validates on the next year. Training samples therefore grow over time, while validation always occurs strictly in the future relative to the corresponding training observations.

Fold	Training period	Validation period	Train n	Validation n
1	2008-07-01–2009-12-31	2010	1,279	1,094
2	2008-07-01–2010-12-31	2011	2,373	951
3	2008-07-01–2011-12-31	2012	3,324	930
4	2008-07-01–2012-11-30	2013	4,254	939
5	2008-07-01–2013-12-31	2014	5,193	1,018
6	2008-07-01–2014-12-31	2015	6,211	734

3.4 Candidate Models and Hyperparameter Selection

Three model families form a purposeful benchmark ladder. **Logistic Regression** provides a regularized linear baseline; **Random Forest** represents a nonlinear bagged-tree ensemble; **Histogram Gradient Boosting** provides a modern boosted nonlinear comparator.

Hyperparameters are selected with **GridSearchCV** using Average Precision as the scoring objective.

3.5 Evaluation Metrics

Because only 24.01% of the analytical sample is positive, accuracy is insufficient by itself. A classifier that always predicts the majority class can appear superficially strong. The study therefore reports accuracy, balanced accuracy, rainfall precision, rainfall recall, rainfall F1, ROC-AUC, Average Precision, and Brier score. **Average Precision** is particularly useful because it summarizes precision-recall performance for the positive class and can be interpreted relative to the underlying positive-class prevalence.

A **DummyClassifier** using the training prior serves as a naïve benchmark. On 2016 validation data it achieved 72.1% accuracy but 0 rainfall recall, 0 rainfall F1, balanced accuracy of 0.50, ROC-AUC of 0.50, and Average Precision of 0.279. This illustrates why accuracy alone is a poor criterion for this task.

3.6 Threshold Selection

The model's probability ranking and the operational classification threshold are treated as separate decisions. The selected model's threshold is optimized using 2016 validation data only. The F1-maximizing threshold was **0.438**. An illustrative asymmetric error rule assigning twice the cost to a missed rainy day produced the same threshold. The 2017 test set played no role in this selection.

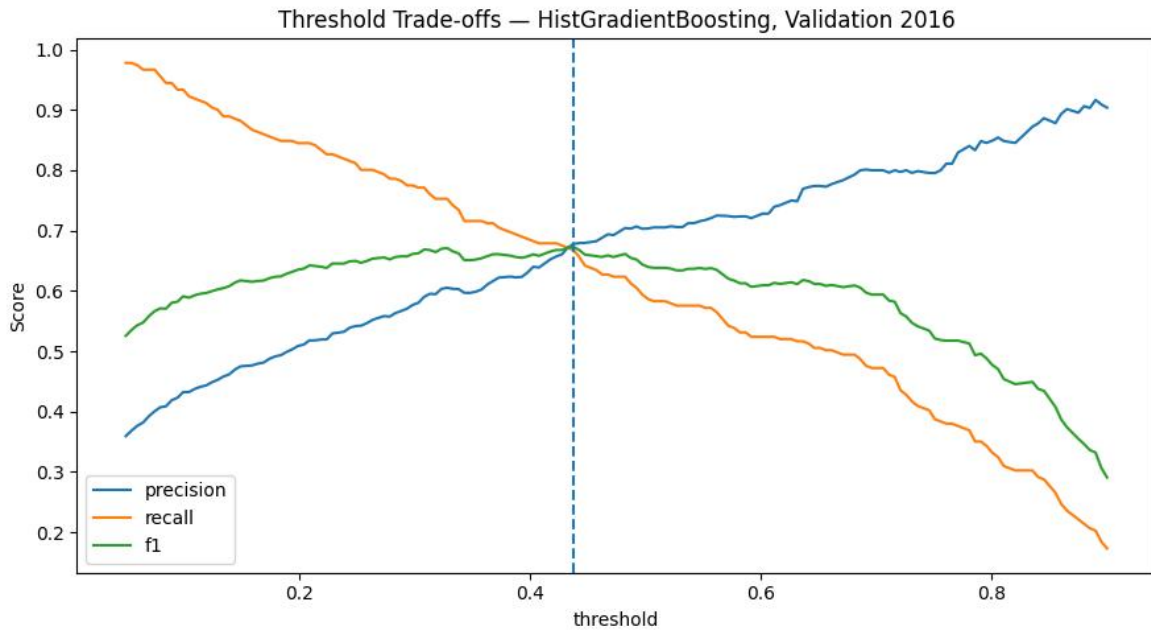


Figure 4. Validation threshold trade-offs for the selected model.

4. Empirical Results

4.1 Validation Model Comparison

Histogram Gradient Boosting produced the strongest 2016 Average Precision (**0.728**) and was selected for final evaluation. Random Forest was close at 0.718, while Logistic Regression achieved 0.689. At the default 0.50 threshold, Histogram Gradient Boosting achieved 81.6% accuracy, 0.746 balanced accuracy, 70.4% rainfall precision, 58.7% rainfall recall, and 0.640 rainfall F1. The narrow differences between the two ensemble models argue against presenting model selection as a dramatic algorithmic victory; the more important contribution is the validation framework.

Model	Accuracy	Balanced acc.	Rain precision	Rain recall	Rain F1	ROC-AUC	Avg. precision	Brier
HistGradientBoosting	0.816	0.746	0.704	0.587	0.640	0.861	0.728	0.130
Random Forest	0.819	0.740	0.727	0.561	0.633	0.860	0.718	0.130
Logistic Regression	0.802	0.742	0.659	0.605	0.631	0.842	0.689	0.138

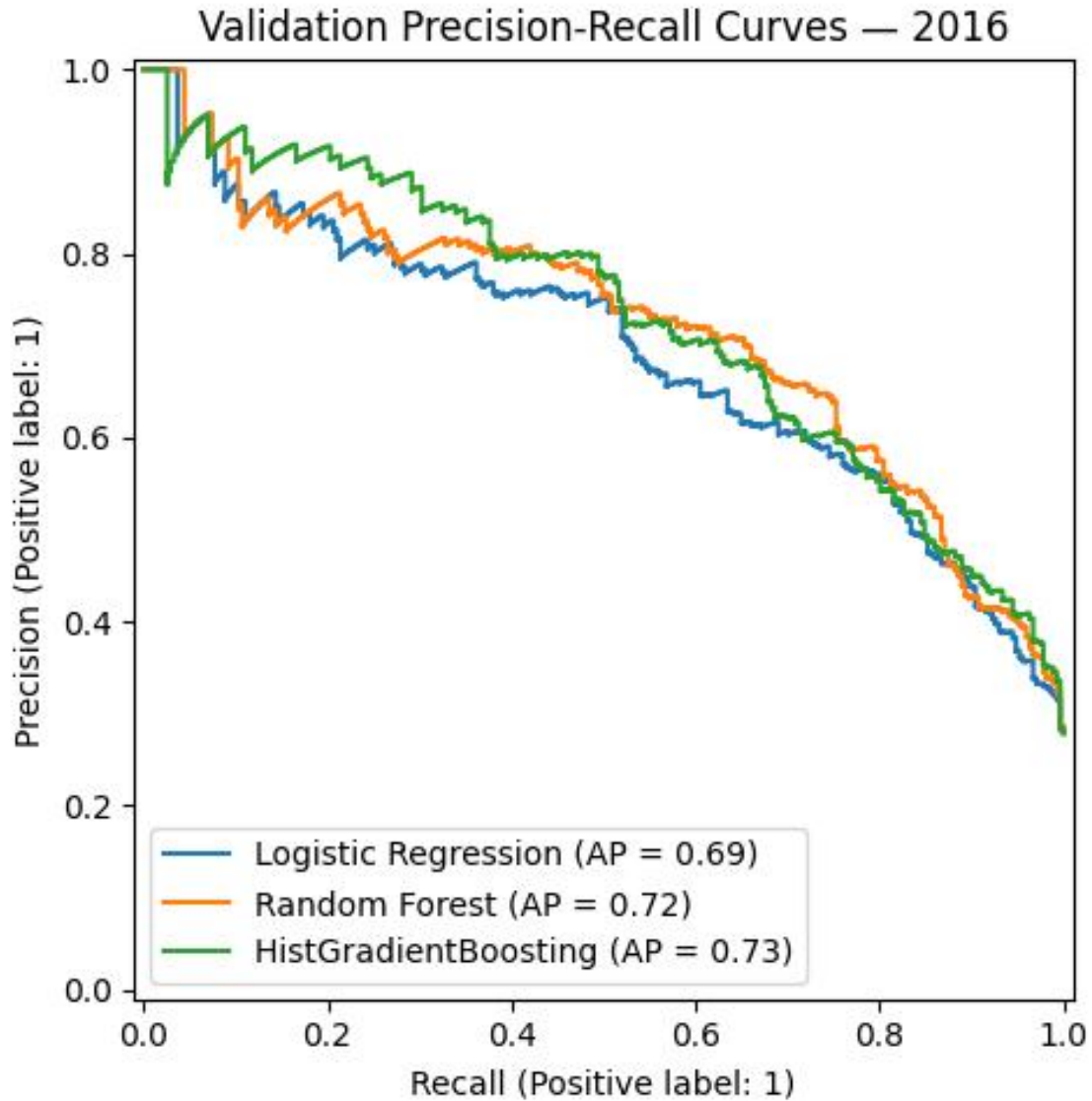


Figure 5. Precision-recall curves for candidate models on the 2016 validation period.

4.2 Final 2017 Out-of-Sample Performance

After model family, hyperparameters, and threshold were fixed, the selected Histogram Gradient Boosting pipeline was refit on all data through 2016 and evaluated once on the 526 observations in 2017. At the validation-selected threshold of 0.438, the model achieved **85.7% accuracy**, 0.676 balanced accuracy, 71.7% rainfall precision, 38.8% rainfall recall, 0.503 rainfall F1, 0.841 ROC-AUC, 0.611 Average Precision, and a Brier score of 0.109.

The threshold adjustment improved rainfall recall from 31.6% at the default 0.50 threshold to 38.8%, while final accuracy increased from 84.8% to 85.7% and rainfall F1 from 0.437 to 0.503. Of 98 rainy observations, 38 were correctly identified and 60 were missed; of 428 dry observations, 413 were correctly classified and 15 were false alarms.

Final test metric	Default 0.50	Selected 0.438
Accuracy	0.848	0.857

Balanced accuracy	0.643	0.676
Rain precision	0.705	0.717
Rain recall	0.316	0.388
Rain F1	0.437	0.503
ROC-AUC	0.841	0.841
Average Precision	0.611	0.611
Brier score	0.109	0.109

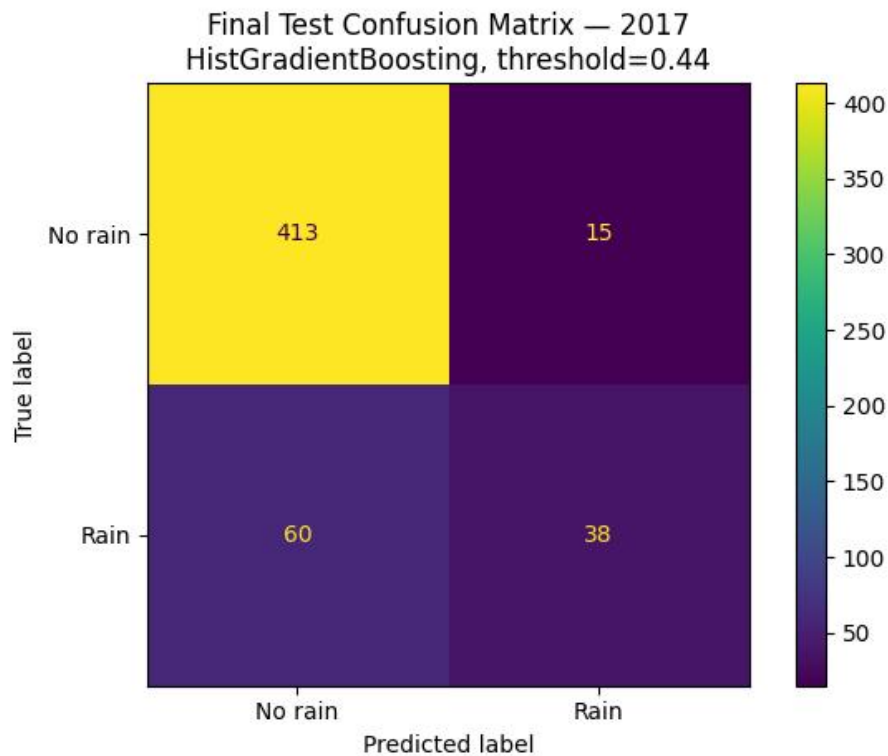


Figure 6. Confusion matrix for the final 2017 test period using the validation-selected threshold.

The decline in rainfall recall from 58.7% in the 2016 validation period to 38.8% in 2017 is a central empirical finding rather than an inconvenient result to hide. The final period had a lower rain prevalence (18.63% versus 27.88% in 2016), and the deterioration demonstrates the difficulty of transporting a threshold across time. The ranking metrics remained comparatively strong—ROC-AUC 0.841 and Average Precision 0.611—suggesting that the model retained useful discriminatory information even though the fixed classification threshold became less sensitive to rainfall.

4.3 Probability Quality and Model Interpretation

The **Brier score** of **0.109** provides a probability-quality measure that penalizes inaccurate probabilistic forecasts. Calibration plots in the technical notebook show how predicted risk aligns with observed event frequency and are important if probabilities are to support decisions rather than merely binary labels.

Permutation importance was computed on the held-out 2017 test set using Average Precision as the scoring function. Scikit-learn describes permutation importance as a model-agnostic procedure that measures the

degradation in predictive performance after a feature is shuffled; when computed on held-out data, it highlights features contributing to generalization rather than training fit.

Pressure3pm produced the largest mean decrease in Average Precision when permuted (0.1487), followed by **Humidity3pm** (0.1111), **WindGustSpeed** (0.0896), and **Sunshine** (0.0694). **Cloud3pm** and **Pressure9am** were materially smaller. These quantities should be interpreted as predictive reliance of the fitted model, not as causal effects on rainfall.

Feature	Mean permutation importance	Std. dev.
Pressure3pm	0.1487	0.0164
Humidity3pm	0.1111	0.0220
WindGustSpeed	0.0896	0.0105
Sunshine	0.0694	0.0182
Cloud3pm	0.0195	0.0062
Pressure9am	0.0178	0.0130
MaxTemp	0.0101	0.0038

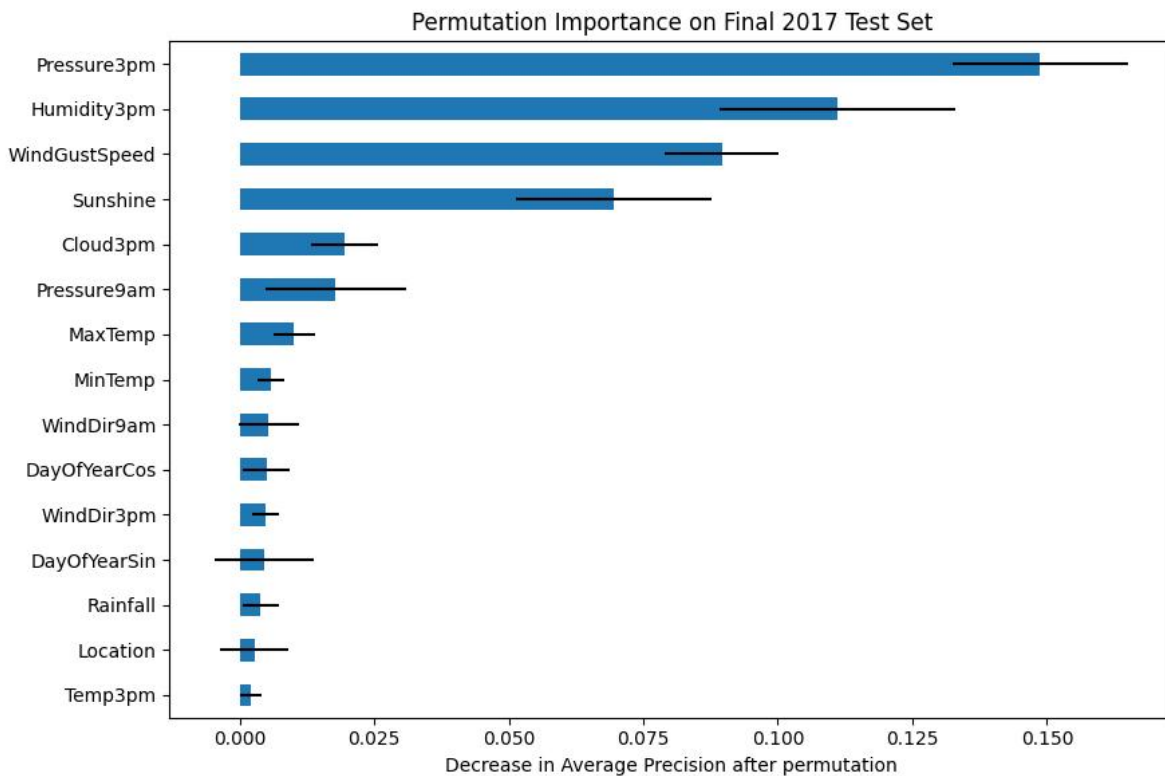


Figure 7. Held-out permutation importance on the 2017 test period.

4.4 Error Heterogeneity

Station-level performance was relatively consistent in the primary 2017 evaluation. Accuracy ranged from 85.2% to 86.4%, rainfall recall from 38.2% to 39.4%, and Average Precision from 0.603 to 0.642 across Melbourne, Melbourne Airport, and Watsonia. This stability suggests that no single station dominated the aggregate result.

Seasonal diagnostics revealed greater heterogeneity. Autumn rainfall recall was 42.9% and summer recall 45.8%, but winter recall was 0 in the 75-observation winter subset, which contained 11 rainy cases. Spring data was absent because the final 2017 test covers 526 observations through 25 June rather than a complete calendar year, and Spring in Australia is between September and November. The model still produced winter ROC-AUC of 0.822 and Average Precision of 0.439, indicating that the probability ranking retained signal while the globally selected threshold failed to convert that signal into positive classifications in this small subgroup. This is a useful warning against treating one fixed threshold as universally optimal.

Season	n	Rain rate	Accuracy	Balanced acc.	Rain recall	Rain F1	Avg. precision
Autumn	274	0.230	0.839	0.695	0.429	0.551	0.682
Summer	177	0.136	0.893	0.710	0.458	0.537	0.542
Winter	75	0.147	0.840	0.492	0.000	0.000	0.439

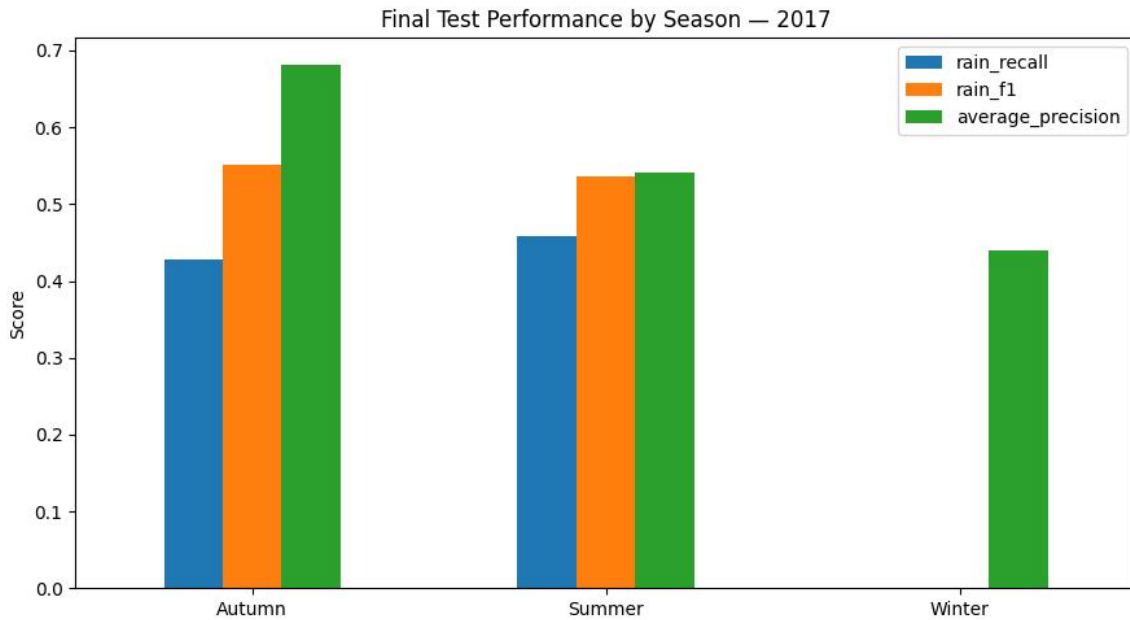


Figure 8. Final-test performance by season.

4.5 Station-Holdout Robustness

To probe spatial dependence, a harder robustness test trained the selected specification on two stations through 2016 and evaluated on the held-out third station in 2017. Average Precision was 0.608 when Melbourne was withheld, 0.632 for Melbourne Airport, and 0.554 for Watsonia. Rainfall recall ranged from 35.3% to 48.4%. The results indicate useful but imperfect transfer across nearby stations and support caution about assuming spatial independence.

5. Discussion

5.1 What the Results Mean

The study finds meaningful predictive signal in prior-day meteorological conditions, but it also shows why predictive performance must be interpreted in context. An 85.7% final accuracy statistic sounds strong in

isolation. Yet the model detected only 38.8% of rainy days in 2017. Conversely, its 0.841 ROC-AUC and 0.611 Average Precision show that the underlying probability ranking remained substantially informative. The practical conclusion is therefore not that rainfall is '85.7% predictable,' but that **the model ranks risk effectively while a fixed decision threshold leaves considerable sensitivity on the table.**

This distinction is central to applied analytics. Model estimation, probability ranking, threshold selection, and business decision rules are related but separate layers. A model can be statistically useful even when a particular threshold is poorly aligned with an operational objective.

5.2 Why Temporal Validation Changes the Interpretation

The original exercise used randomized stratified splitting and reported approximately 84% Random Forest accuracy with roughly 51% rainfall recall. Those results were appropriate as evidence of classification workflow, but random splitting mixes earlier and later dates. The redesigned study asks the harder question: **can a model trained on the past generalize to a later period?** Scikit-learn explicitly warns that classical cross-validation can produce inappropriate generalization estimates for **time-dependent** observations because nearby observations are correlated and future data can enter training.

The final temporal results therefore should not be compared mechanically with the original 84% as if they were evaluations of the same experiment. The portfolio contribution is the **redesign itself: the validation protocol now better matches the claimed forecasting use case.**

5.3 Relevance to Business and Data Analytics

Although the substantive domain is weather, the analytical architecture is directly transferable to **business prediction problems**. Customer churn, loan default, fraud, demand, supply disruption, equipment failure, and marketing response all involve the same core questions: What information is genuinely available at decision time? How should missing values and categories be processed without leakage? Which validation period resembles deployment? Which error matters more? Should the model optimize ranking or a binary action? Does performance deteriorate across time or segments?

The project therefore demonstrates business-analytics competencies beyond model fitting. It connects technical modeling to decision design: threshold choice can reflect asymmetric error costs; calibration matters when probabilities drive resource allocation; subgroup diagnostics reveal where a policy may fail; and temporal holdouts provide evidence about whether a model can survive distribution change.

5.4 Skills Demonstrated

- **Python data analysis** with pandas, NumPy, and Matplotlib.
- Reusable **scikit-learn Pipeline** and **ColumnTransformer** architecture for mixed data types.
- Leakage-aware imputation, scaling, and categorical encoding.
- **Supervised binary classification** with linear, bagged-tree, and boosted-tree models.
- **GridSearchCV** with expanding temporal folds and a class-appropriate optimization metric.
- **Evaluation** with confusion matrices, precision-recall, ROC, calibration, Brier score, and subgroup diagnostics.
- Validation-only threshold selection and asymmetric error-cost reasoning.
- Model-agnostic permutation importance on held-out observations.
- Critical interpretation of model deterioration rather than selective reporting of favorable metrics.

6. Limitations and Future Research

First, the dataset is observational. Feature importance measures predictive reliance and cannot establish that atmospheric pressure, humidity, wind, or sunshine causally determine the model outcome.

Second, deployment timing requires care. Bureau of Meteorology rainfall observations are conventionally tied to 9 a.m. measurement windows, while the dataset also contains 3 p.m. variables. A production forecasting system must define exactly when a prediction is issued and include only measurements available at that moment.

Third, the primary model is geographically local. The Melbourne-metro focus improves climatic coherence but limits external validity. A natural extension is to compare localized models with a nationwide hierarchical or location-aware model.

Fourth, nearby stations can observe the same weather systems, so residual spatial dependence remains possible. The station-holdout exercise provides a useful robustness check but does not explicitly model spatial correlation.

Fifth, the final 2017 test covers 526 observations through 25 June rather than a complete calendar year. The winter subgroup is particularly small. Seasonal conclusions should therefore be treated as diagnostics, not stable population estimates.

Sixth, median/mode imputation is transparent and leakage-safe but does not model the missingness mechanism. Future work could compare missing-indicator strategies or more sophisticated imputation under the same temporal protocol.

Seventh, the F1-maximizing threshold is a statistical compromise rather than an economically optimized policy. A business deployment would estimate the relative cost of missed events and false alarms and select thresholds from those decision consequences.

Finally, temporal performance deterioration suggests a need for monitoring and retraining. Future work could implement rolling-origin evaluation, drift diagnostics, recalibration, and threshold re-estimation as new periods become available.

7. Conclusion

This portfolio project demonstrates an end-to-end **machine-learning** study in which methodological design is treated as seriously as algorithm choice. Starting from an IBM certificate exercise, the analysis was redeveloped around an explicit **next-day forecasting question, leakage-safe preprocessing, chronological validation, imbalanced-class evaluation, model comparison, threshold analysis, probability calibration, held-out interpretation, and subgroup robustness.**

Histogram Gradient Boosting was the strongest validation model and achieved **0.841 ROC-AUC** and **0.611 Average Precision** on the untouched 2017 test period. At the validation-selected threshold, it reached **85.7% accuracy** and 71.7% rainfall precision but only 38.8% rainfall recall. The divergence between ranking quality and threshold sensitivity, together with the seasonal error analysis, is the most important substantive lesson: responsible predictive analytics requires understanding how validation design, prevalence, thresholds, and decision costs shape the meaning of model performance.

The project is intended to demonstrate progression from technical execution to research-oriented reasoning: not only how to build and tune machine-learning models in Python, but how to question the validity of an evaluation, redesign an empirical workflow, interpret imperfect out-of-sample evidence, and connect predictive outputs to decisions.

8. Project Provenance and Reproducibility

The precursor project was completed within the IBM Machine Learning Certificate. The original notebook used complete-case deletion, Melbourne-area filtering, a season feature, randomized stratified train/test splitting, ColumnTransformer/Pipeline preprocessing, GridSearchCV, Random Forest and Logistic Regression, classification metrics, and Random Forest feature importance.

The present portfolio redevelopment preserves that technical foundation while adding the methodological extensions described in this report. The companion Jupyter notebook is fully executed and contains the complete code, saved outputs, figures, hyperparameter searches, and diagnostic analyses underlying every numerical result reported here. This report is intentionally interpretive rather than a cell-by-cell reproduction of the notebook.

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Appendix A. Selected Numerical Results

Analysis	Key result
Analytical sample	8,443 observations; 24.01% positive class
2016 validation winner	Histogram Gradient Boosting; Average Precision 0.728
Selected threshold	0.438, chosen on 2016 validation data
2017 final accuracy	0.857
2017 balanced accuracy	0.676
2017 rain precision / recall / F1	0.717 / 0.388 / 0.503
2017 ROC-AUC / Average Precision	0.841 / 0.611
2017 Brier score	0.109
Top permutation feature	Pressure3pm; mean importance 0.1487
Station-holdout AP range	0.554–0.632

Appendix B. Companion Technical Artifact

Notebook title: Predicting Daily Rainfall in Melbourne Using Machine Learning: A Temporal Validation and Model Comparison Study.

The fully executed notebook contains the complete Python implementation, preprocessing pipelines, expanding-year cross-validation, model searches, diagnostic plots, threshold analysis, calibration analysis, permutation importance, station/season diagnostics, and station-holdout robustness tests. The notebook was submitted or linked alongside this report so that reviewers can inspect the reproducible technical evidence behind the research narrative.